

NOVEL BOILING-ENHANCED MULTI-CHANNEL HEAT SINKS

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ABSTRACT

This paper describes the results of experimental investigation of heat transfer characteristics of the novel type of multi-channel heat sinks with micro- and meso-sized channels fabricated using a newly developed polycapillary materials (PCM) technology.

The enhancement of heat transfer properties is achieved by a combination of both structural and flow parameters under a single- or a two-phase regime. The proposed technology enables miniature heat sinks with arrays of longitudinal small-diameter (down to tens of microns) channels, produced of variety of materials including metals, alloys and ceramics, resulting in high surface density. The latter ensures superior volumetric heat transfer coefficients under condition of forced single-phase convection of fluid. However, since the heat transfer during a single-phase regime is still limited by significant thermal and hydraulic resistances, its further enhancement is associated with a phase transition and a resulting two-phase flow. The latter allows for much higher heat transfer coefficients and lower pressure drop due to steam formation.

The objective of this paper is to demonstrate the potential of using the proposed multi-channel heat sinks for variety of applications requiring withdrawal of significant amounts of heat within small volume.

As a result of conducted research, preliminary heat transfer characteristics of the proposed heat sinks were evaluated and analyzed. It was found that heat transfer coefficients observed during a single-phase regime are in good agreement with theoretical equations describing forced laminar convection in channels. When combined with high surface density, they demonstrate superior volumetric ability to reject up to $6 \times 10^6 \text{ W/m}^3\text{-K}$ of heat under moderate flow rates.

By providing sufficient superheat, a phase transition (boiling) can be induced resulting in 2.3 – 2.6 times

increase in rejected heat flux. This result is in part due to the unique fabrication process allowing producing flow channels with roughened internal surface formed by sintered particles, which serve as perfect nucleation sites.

In the future, additional study will be conducted analyze in detail thermal and hydraulic properties as well as specific designs and material options of the proposed boiling-enhanced multi-channel heat sinks.

NOMENCLATURE

d	– hydraulic diameter of mini-channels, m
N	– number of channels
C	– channel fraction in the cross-section
A_b	– surface area of the base, m^2
A	– surface area of the channels, m^2
K	– surface density of the channels, m^2/m^3
Q	– amount of heat transferred, W
T_w	– temperature of external surface of the sink, $^\circ\text{C}$
$T_{c,ni}$	– inlet temperature of the water, $^\circ\text{C}$
$T_{c,out}$	– outlet temperature of the water, $^\circ\text{C}$
ΔT_{SS}	– temperature drop on a SS plate
G_c	– mass flow rate of air, kg/s
U	– overall heat transfer coefficient, $\text{W}/(\text{m}^2\text{ }^\circ\text{C})$
S_c	– cross-section area of the heat sink, m^2
Nu	– Nusselt number
Re	– Reynolds number
Pe	– Peclet number
Pr	– Prandtl number
a	– sample length, mm
b	– sample width, mm
t	– sample thickness, mm
w	– air velocity in mini-channels, m/s
k_c	– coefficient of thermal conductivity of water, $\text{W}/(\text{m }^\circ\text{C})$

- k_{SS} – coefficient of thermal conductivity of SS plate, $W/(m^{\circ}C)$
 x_{SS} – SS plate thickness, m
 ρ – density of water, kg/m^3
 ν – kinematic viscosity of water, m^2/s

INTRODUCTION

Rapid development of the next generation, high density and high-speed microelectronic circuits and computers requires implementation of new methods and materials capable of providing effective cooling within restricted volume. Recently introduced micro- and meso-scale heat sinks fabricated using various technologies appeared to offer a potential solution to this problem by allowing dramatic reduction in thermal resistance on the coolant side resulting from a small channel diameter and high wetted surface area. More than a 60 times increase in the overall heat transfer coefficient of micro heat sinks as compared to that of conventional macro-scale heat exchangers was reported [1]. At the same time, high pressure drops may be required to provide such cooling capabilities. The issue of hydraulic resistance becomes especially important when liquid coolants are utilized in attempt to increase heat dissipation and thus minimize the maximum substrate temperature as well as reduce the substrate temperature gradients. For example, current dense packaging of high power electronics demands as much as 100 W/cm^2 heat dissipation at chip levels while maintaining the device at temperatures below $85^{\circ}C$ [2].

Further enhancement of micro-scale heat sinks is associated with phase transition inside the channels, which is characterized by significantly higher heat transfer coefficients as compared to a single-phase forced convection and reduced pressure drop resulting from a two-phase flow.

In order to design an effective micro-scale heat sink, it is necessary to know parameters, such as pressure required to pump the coolant, its flow rate, heat transfer coefficient and temperature distributions. To obtain these characteristics, separate study is needed for each type of heat sink since the experimental studies reported in the literature showed that many phenomena in micro-scale flow and heat transfer differ from those in macro-channels, specifically in the area of boiling and two-phase flows. A comprehensive review of these investigations conducted over the past decade is presented in [3, 4, 5].

This paper presents the results of initial experimental study of the heat transfer characteristics of boiling-enhanced micro- and meso-channel heat sinks fabricated using the polycapillary materials (PCM) technology recently developed at MER Corporation [6]. These materials are characterized by highly extended inner surfaces, formed by a system of parallel through passages with hydraulic diameter varying from several microns to several millimeters resulting in surface density of $10,000\text{ m}^2/\text{m}^3$ and higher. High surface area so achieved at relatively small channel size is a key advantage of this technology enabling various applications of such materials in actively cooled thermal management systems. In addition, possibility of producing PCM with porous

inter-channel walls allows extending the application area to include transpiration and convection-transpiration (*contran*) cooling options [7]. To provide higher thermal conductivity, PCM could be fabricated out of numerous high thermal conductivity materials, such as: copper, aluminum, their alloys; other materials, like nickel, titanium, stainless steel; or composites including aluminum- and copper-carbon, copper-cubic boron nitride, etc. By using commonly available tools and equipment, the process can be fully automated to provide high production rates at reasonable cost.

Presented below are the results of experimental thermal and hydraulic characterization of selected multi-channel samples fabricated using PCM technology, which provide important relationships and expressions necessary to establish their potential for specific practical applications.

EXPERIMENTAL

A series of multi-channel samples was fabricated using copper with various geometric characteristics shown in Table 1.

Table 1. Geometrical characteristics of tested PCMs

Sample ID	a x b x t mm	d, mm	N	C, %	K, m^2/m^3
1	29x34x6.4	1.8	24	33	730
2	36x30x5.8	1.1	28	13	463
3	29x22x4.2	0.6	141	34	2181
4	23x28x3.9	0.08	2214	11	6200

The schematic and the general view of the produced samples are shown in Fig. 1 and 2 below.

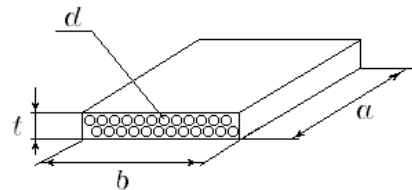


Fig. 1 Schematic of the PCM-based heat sink

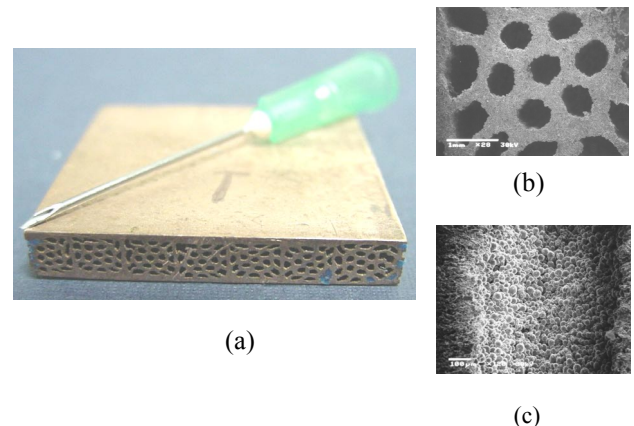


Fig. 2 Heat sink sample (a), SEMs of the cross-section (b) and the structure of internal channel surface (c)

By selecting the samples as shown in Table 1, the objective was to evaluate the effect of channel size,

number of channels and the roughness of the internal channel walls on overall thermal performance of the experimental heat sink including heat transfer coefficient and phase transition.

To realize a forced convection heat transfer, the experimental setup was constructed as shown in Fig. 2

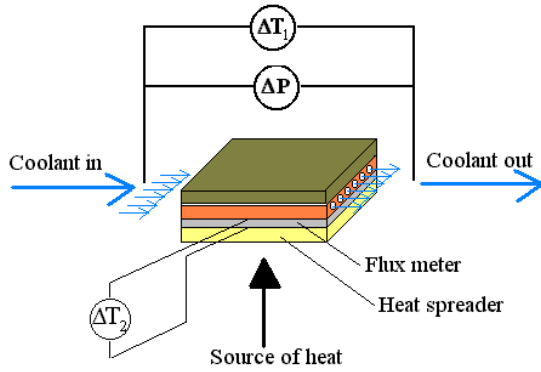


Fig. 3 Schematic of experimental setup

The experimental setup consisted of a heat sink sample with two manifolds providing the inlet and outlet of a coolant as well as necessary ports for measuring the temperature and pressure drops across the sample. The mass flow rate of the coolant (water) was controlled by means of a rotameter-type flow meter. To increase accuracy of flow rate readings, several flow meters covering different ranges of flow rates were utilized. A propane torch was used as a heat source in order to achieve high incident heat fluxes (up to 150 W/cm²). The heat was first supplied to the heat spreader (a copper plate) and delivered to the sample through a thin stainless steel plate serving as a heat flux gauge. By measuring the temperature of both sides of this plate of known thickness, the value of the heat flux passing through the sample can be calculated. The overall heat transfer coefficient from hot surface to cooling water inside the channel was then calculated using the temperature difference between the sample wall and the coolant as follows [8]:

$$U = \frac{Q}{A \Delta T_{w-c}}, \quad (1)$$

where ΔT_{w-c} is a log mean temperature difference, defined as:

$$\Delta T_{w-c} = \frac{(T_w - T_{c,in}) - (T_w - T_{c,out})}{\ln \frac{T_w - T_{c,in}}{T_w - T_{c,out}}} = \frac{T_{c,out} - T_{c,in}}{\ln \frac{T_w - T_{c,in}}{T_w - T_{c,out}}} \quad (2)$$

In expression (1), the calculated heat transfer coefficient is based on the wetted area defined as:

$$A = \pi d a N \quad (3)$$

The heat flux Q used in expression (1) can be obtained from either of the following equations representing the heat load balance in the heat sink:

$$Q = k_{ss} A_b \frac{\Delta T_{ss}}{x_{ss}} \quad (4)$$

$$Q = G_c C_p (T_{c,out} - T_{c,in}) \quad (5)$$

In case of boiling, expression (4) has to be used since (5) does not reflect the starting point of the phase transition and, thus, is not adequate.

The dimensionless characteristic numbers can then be determined as follows:

Reynolds Number:

$$Re = \frac{\rho w d}{\mu} = \frac{G_c d}{S_c \mu} \quad (6)$$

Peclet Number:

$$Pe = Re Pr \quad (7)$$

Nusselt Number:

$$Nu = \frac{U d}{k_c} \quad (8)$$

Graetz Number:

$$Gz = \frac{Pe d}{a} \quad (9)$$

The latter was used to characterize heat transfer in PCM heat sinks while accounting for the entrance region, since in most cases the flow in small channels is not fully developed [5].

In addition, a volumetric heat transfer coefficient was determined as follows:

$$U_v = U K \quad (10)$$

to compare the heat transfer capability per unit volume and, thus, characterize compactness of the tested heat sink samples.

RESULTS AND DISCUSSION

As a result of experimental evaluation, a set of data points was obtained characterizing the heat transfer properties of the described multi-channel heat sinks under condition of a single-phase and a two-phase flow.

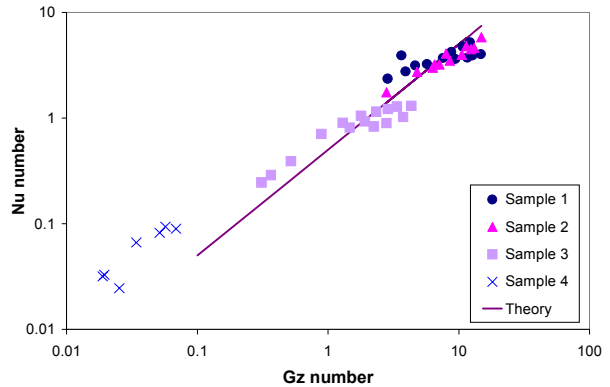


Fig. 4 Experimental relationship $Nu = f(Gz)$ for a single-phase flow

The data presented in this graph relate to a single-phase flow, where heat transfer occurs due to forced convection of water. As can be seen from the figure, the experimental points for all the samples line-up closely around the theoretical line $Nu = 0.5Gz$ resulting from a theoretical solution for a laminar heat transfer under condition of low Re numbers and large l/d ratios [9].

Since all the samples have different surface density, the amount of heat rejected from a unit volume under similar flow rates of a coolant will vary. As shown in Fig. 5 below, higher surface density results in higher values of volumetric heat transfer coefficients.

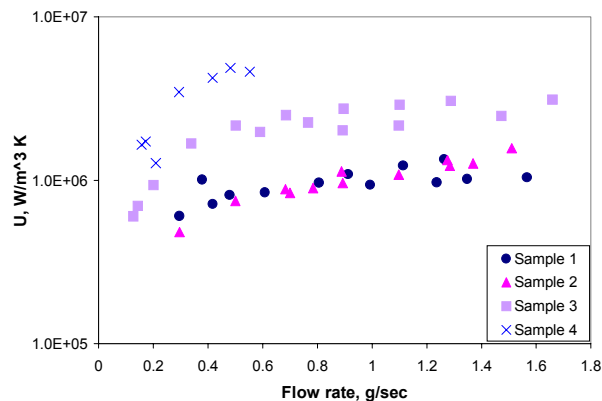


Fig. 5 Volumetric heat transfer coefficients vs flow rate

Naturally, the samples with higher surface density and smaller channels will demonstrate higher hydraulic resistance. Therefore, careful consideration should be given to both aspects when selecting a suitable heat sink for a specific practical application.

One of the methods allowing further increase in the amount of heat rejected by a heat sink while lowering its hydraulic resistance is associated with phase transition. Therefore, described multi-channel samples were also tested under condition of two-phase flow by providing sufficient superheat to the heated surface. The beginning of phase transition (boiling) was associated with portions of steam coming out of channels. The summary of the experimental data obtained in this case is presented in Table 2 in comparison with results obtained for the single-phase flow.

Table 2. Heat flux increase due to phase transition

Sample	G, g/sec	Q_1/Q_2
1	0.3	2.00
	0.5	1.66
	0.6	1.39
2	0.3	2.64
	0.5	1.98
	0.7	2.19
3	0.2	2.20
	0.3	2.19
	0.5	1.90
4	0.1	2.35
	0.2	2.17
	0.3	1.40

As can be seen from the Table, phase transition due to surface superheat results in considerable (up to 2.64 times) increase in the amount of heat rejected by the heat sink samples. The level of increase is dictated by a boiling regime inside the channels, which corresponds to availability and a number of nucleation sites on the channel surface. By comparing results obtained for samples 1 and 2 with a smooth machined surface and a rough sintered particulate surface of channels (Fig. 2c) respectively, one can notice that increase in heat flux as a result of phase transition is more pronounced in case of sample 2 which obviously has more nucleation sites. At the same flow rates and similar wall temperatures, channels with rougher surface generate more bubbles resulting in more than 30% higher heat fluxes as compared to smooth-surface channels. This percentage increases to almost 65% provided the superheat is increased by 10 degrees. By comparing multi-channel samples 3 and 4, similar increase in heat flux level can be observed. However, at increased flow rates, the flow velocity in micro-channels becomes so high that bubble formation is ceased resulting in lower heat flux values.

Due to limited capacity of the heating source utilized in these experiments, the maximum heat flux applied to the heat sink samples was between 65 and 70 W/cm². Higher heat loads will be investigated during future research activities.

CONCLUSIONS

Based on experimental investigation of heat transfer properties of the advanced multi-channel heat sinks, the following conclusions can be made.

1. A single phase forced laminar convection of water inside various-size micro- and meso-scale channels results in heat transfer coefficients close to those predicted by a theoretical solution.
2. High surface area, easily achievable in the described heat sink samples, and thus, high surface density result in incredibly high volumetric heat transfer coefficients even at lower flow rates.
3. Facilitating a phase transition by providing sufficient superheat to the tested heat sinks further enhances their ability to remove 30 to 60% more heat as compared to a single-phase forced convection under same mass flow rates.
4. Proposed fabrication technique provides technological feasibility of the described multi-channel heat sinks by enhancing phase transition via internal channel surface structure formed by sintered metal/ceramic particles.

During the future research effort, the authors will deeper investigate the phase transition process enhancement using this technology as well as extend the obtained results over a broader range of heat sink materials and designs.

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